

Bayesian Methods for density Estimation.

Recall, in ML estimate the parameter set (θ) is considered deterministic.

When n is small, ML estimate may not be optimal.

Eg: Consider ML estimate for a Ber. RV.

$$\theta_{ML} = \frac{\text{no. of times the trial has yielded 1}}{n}$$

suppose $D = \{1, 1, 1, 1\}$

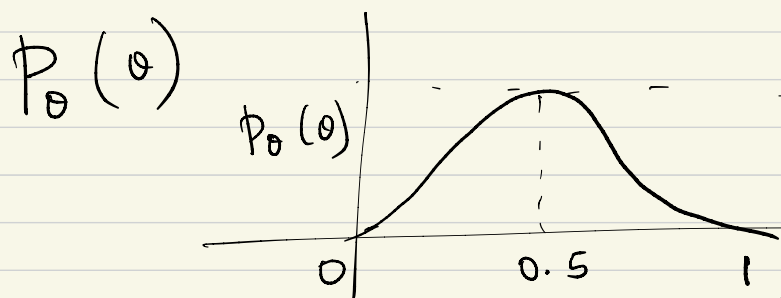
$$\theta_{ML} = 1$$

The above estimate is not "optimal" if the true value is "around 0.5".

Bayesian Estimation: Incorporate the prior belief / information about the param.

Assume the parameter θ to be a Rv
& model the prior belief in its distribution.

In the previous example for Ber, Rv.



Suppose $x \sim p_x$

$\theta \sim p_\theta$: prior on param.

compute

$$p_{\theta|x}(\cdot) \propto p_\theta(\theta) \cdot p_\theta(x|\theta)$$

How to use $p_{\theta|x}$?

MAP estimate : Take the mode
of $p_{\theta|x}$ as the estimate for θ .

Conjugate prior : The prior on θ , that ensures the posterior to have the same distributional form as itself.

Example : MAP Estimate for Bernoulli

$$x \sim \text{Ber}(\theta)$$

$$P_{x|\theta} = \theta^x \cdot (1-\theta)^{1-x}$$

$$P_{\theta}(\theta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \cdot \Gamma(\beta)} \cdot \theta^{\alpha-1} \cdot (1-\theta)^{\beta-1}$$

where $\Gamma(\)$ is the std. gamma fn.
& α, β are user-chosen params.

Compute $P_{\theta|x} \propto P_{x|\theta} \cdot P_{\theta}$

$$P_{\theta|x} \propto \left[\theta^x \cdot (1-\theta)^{1-x} \right] \cdot \theta^{\alpha-1} \cdot (1-\theta)^{\beta-1}$$

$$\propto \theta^{x+\alpha-1} \cdot (1-\theta)^{\beta-x}$$

In MAP estimate, θ_{MAP} is taken to be the mode of $p_{\theta|x}$

In the above case,

$$\theta_{\text{MAP}} = \frac{\sum_{i=1}^n x_i + \alpha - 1}{n + \alpha + \beta - 2}$$

$$\theta_{\text{ML}} = \frac{\sum_{i=1}^n x_i}{n}$$

Non-parametric Density Estimation.

These methods can evaluate the density function at a point w/o making any parametric form assumption.

Suppose p_x is the density fn. that is being evaluated.

$$x \in \mathbb{R}^d,$$

Let R denote a "region" on the support of p_x .

Let P be the probability of a point x "falling" within the region R .

$$P = \int_R p_x(x) dx$$

Suppose we have n datapoints drawn from p_x (iid).

If the event of a datapoint falling within the region R , be modeled as a Bern. RV,

the Max likelihood estimate for the parameter is given by

$$P = \frac{k}{n},$$

where k is the no of points within the region.

from earlier,

$$\frac{k}{n} = \int_R p_x(x) dx$$

If R is small enough,

$$\frac{k}{n} = p_x(x) \cdot V \quad \text{where}$$

V is the "volume" of the region.

$$\Rightarrow p_x(x) = \frac{k}{n \cdot V}$$

Two ways to construct NP density estimator,
fix V & count k
fix k & grow V to encompass k points.

Parzen Window Estimator

Fix V & count k , to evaluate density.

suppose we want to evaluate the density fn at a point x .

let R be a d -dimensional hypercube around x with length h .

$$V = h^d$$

To count k , define a window fn

$$\phi(u) = \begin{cases} 1 & \text{if } |u_j| \leq \frac{1}{2} \quad j=1 \dots d \\ 0 & \text{otherwise} \end{cases}$$

$\phi\left(\frac{x - x_i}{h}\right)$: Indicator whether x_i is within the hypercube of volume h^d centered around x

$$p_x(x) = \frac{k}{n \cdot V}$$

$$p_x(x) = \frac{\sum_{i=1}^n \phi\left(\frac{x - x_i}{h}\right)}{n \cdot h^d}$$

$\phi(u)$ can be made smoother by

choosing $\phi(u) = \exp\left(-\|u - u_0\|_2^2\right)$

Nearest Neighbour Estimator

non-parametric Estimate where
fix K & grow V .

$$P_X(x) = \frac{K}{nV}.$$

Example use case : m -class classification.

Bayes Classifier with Nearest Neighbour
density Estimate for $P_{Y|X}$.

Suppose we want to estimate the
class of a sample x .

we place a volume V around x &
capture K -samples.

Let K_i be the number of points
within the volume V corresponding
to the i th class.

$$\sum_{i=1}^m K_i = K$$

NN estimate for $p(x, y_i) = \frac{k_i}{nv}$

To construct a Baye's classifier

$$p(y_i|x) = \frac{p(x, y_i)}{\sum_y p(x, y_i)}$$

$$= \frac{k_i/nv}{\sum_{j=1}^M \frac{k_j}{nv}}$$

$$p(y_i|x) = \frac{k_i}{K}$$

$$h_B(x) = i \quad \text{if} \quad p(y_i|x) > p(y_j|x)$$

$$= i \quad \text{if} \quad \frac{k_i}{K} > \frac{k_j}{K}$$

$$= i \quad \text{if} \quad k_i > k_j$$

k-nearest neighbour classifier.